

Standard Operating Procedure

Active Directory Time Synchronization — corp.lab

Objective: Resolve “time change was too big” synchronization errors on domain-joined endpoints, and configure the Primary Domain Controller (PDC Emulator) to pull from a reliable external NTP source, overriding the local CMOS clock or hypervisor-injected time.

Part 1: Resolving Client-Side Synchronization Failures

Symptom

When attempting to force a time sync on a domain client, the system returns:

```
The computer did not resync because the required time change was too big.
```

Cause

The time difference between the local machine and the Domain Controller exceeds the maximum allowed phase correction threshold (typically 15–48 hours). The Windows Time Service automatically rejects large time jumps as a security precaution to protect Active Directory replication and Kerberos authentication.

Precondition: Disable Hyper-V Time Synchronization (if virtualized)

If the client is a Hyper-V VM, the Time Synchronization integration service competes with the domain w32time service and will silently re-inject host time, undoing the manual bridge below. Check and disable it from the Hyper-V host **before** proceeding.

Run on the Hyper-V host:

```
Get-VMIntegrationService -VMName "<VMNAME>" | Where-Object {$_.Name -eq "Time Synchronization"}

# If Enabled : True, disable it:
Disable-VMIntegrationService -VMName "<VMNAME>" -Name "Time Synchronization"
```

Resolution: Manual Bridge and Resync

Once Hyper-V time sync is confirmed disabled (or not applicable), manually bump the client's local clock to match the Domain Controller's time (within a 5-minute tolerance window) before forcing synchronization.

1. Check the current time on the Domain Controller:

```
net time \\DC01.corp.lab
```

2. Manually set the local client clock to match the DC's time (adjust for local timezone, e.g. IST):

```
Set-Date -Date "6/29/2026 7:12:00 PM"
```

3. Force the Windows Time Service to resync:

```
w32tm /resync /force
```

4. Verify the client is now successfully tracking the Domain Controller:

```
w32tm /query /status
```

Expected: Source reads DC01.corp.lab and Last Successful Sync Time is current.

Part 2: Configuring the Domain Controller (PDC Emulator) Time Source

Symptom

When querying time status on the Domain Controller (DC01), the output shows:

```
Stratum: 1 (primary reference - syncd by radio clock)
Source: Local CMOS Clock (or VM IC Time Synchronization Provider)
```

Cause

The DC is relying on its local hardware clock (or host hypervisor) rather than an authoritative NTP server. Hardware clocks drift, and this drift is broadcast to the entire domain via the PDC Emulator.

Resolution: Disable Local/VM Clocks and Enforce NTP

Configure the PDC Emulator to strictly use reliable external NTP servers (e.g. regional pools such as in.pool.ntp.org) and advertise itself as a reliable source to the domain. Run the following in an elevated PowerShell prompt on DC01.

1. Force the Windows Time Service to operate as an NTP client:

```
Set-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Services\W32Time\Parameters"
-Name "Type" -Value "NTP"
```

2. Disable Virtual Machine Time Injection (if virtualized via Hyper-V / Proxmox / ESXi):

Prevents the hypervisor's integration tools from overriding the NTP configuration with the host's hardware clock.

```
Set-ItemProperty -Path
"HKLM:\SYSTEM\CurrentControlSet\Services\W32Time\TimeProviders\VMICTimeProvider" -Name
"Enabled" -Value 0
```

3. Configure the external NTP peers and announce flags:

The 0x8 flag ensures the service sends regular client requests required by most public NTP pools.

```
w32tm /config /syncfromflags:manual /manualpeerlist:"0.in.pool.ntp.org,0x8
1.in.pool.ntp.org,0x8 2.in.pool.ntp.org,0x8" /reliable:yes /update
```

4. Restart the Time Service and force rediscovery:

Clears the old CMOS/hypervisor time source data and forces a fresh synchronization with the external servers.

```
Restart-Service w32time
w32tm /resync /rediscover
```

Verification

Confirm the new configuration has been applied successfully:

```
w32tm /query /status
```

Source should display one of the configured NTP servers (e.g. 0.in.pool.ntp.org).

Stratum should read 2 or 3, indicating synchronization with a primary network time server.

Note: Apply Part 1's Hyper-V precondition check to any domain-joined VM before manual time bridging. Order matters — reconfigure syncfromflags before issuing a resync to avoid regression to Stratum 0 / Local CMOS Clock.